Software Quality Assurance

Important Short Questions (15) · Topics not covered in the long questions

Learn first (★★★)	Then read once (★★)
Q1 Defect · Q2 Handling a defect · Q6 Process vs Product · Q7 Internal vs External · Q8 Quality metric · Q9 Correctness vs Reliability	Q3, Q4, Q5, Q10, Q11, Q12, Q13, Q14, Q15

Topics already in the long questions (QA vs QC, Cost of Quality, Benchmarking, Garvin, Dimensions, McCall/ISO, Basic/Expected/Exciting) are not repeated here. If one of them comes as a short, write 3–4 points from its long-question table.

PPT 1: Quality

Q1. What is a defect? ★★★ (Slide 19)

A defect is the result of a **mistake** made during the development process. Defects introduce **faults** in the software. Defects can also be **missing, ambiguous or incorrect requirements**, which lead to software that does not match the needs of its users.

Sequence: Error → Defect → Fault → Failure

Example: a developer misunderstands a requirement (error), writes wrong code (defect/fault), and the software gives a wrong result to the user (failure).

Q2. What is the best way to handle a defect? ★★★ (Slide 20, 22)

The most cost-effective way to handle a defect is to **prevent** it. Every defect that is prevented removes the **rework** and effort needed to fix it.

The **shorter** the time between introducing and fixing a defect, the **less** rework is needed. A requirements defect that costs **1 unit** to fix in the requirements phase can cost **100+ units** if it is found after delivery, because the cost grows **exponentially**.

Q3. What is quality? ★★ (Slide 3–4, 10)

There is **no single definition** of quality; different organizations define it differently. Generally, quality means the **desirable characteristics** a product should have.

- Quality means **fitness for use**.
- Quality is **inversely proportional to variability**.
- **ISO definition:** quality is the degree to which a system **meets specified requirements** and customer or user needs and expectations.

Q4. Differentiate between quality of design and quality of conformance. ★★ (Slide 5)

Quality of Design	Quality of Conformance
Products are made in different grades or levels **on purpose**	How well the product **matches its specification**
Variations are **intentional**	Depends on the manufacturing process, training, supervision, tests and inspections
Example: cheap and expensive phones or cars	Example: a phone works exactly as its design says

Q5. What are the benefits of software quality? ★★ (Slide 17–18)

- **Fewer defects**, and defects are prevented early at less cost
- **Time saving**, because less rework shortens development time
- **Cost saving**, because fewer failures happen after delivery
- **Higher reliability** and lower maintenance cost
- **Meets the actual needs** of the customer
- **Higher satisfaction** of both users and developers

PPT 2: Software Quality Attributes

Q6. Differentiate between process quality and product quality. ★★★ (Slide 4, 6)

Process Quality	Product Quality
Measure of the **activity** that produces a product	Measure of the **product itself**
Focuses on **how** the software is made	Focuses on **what** is delivered
Example: productivity, timeliness	Example: final software, specifications, models, plans

The underlying assumption is that a **quality process leads to a quality product** (CMMI).

Q7. Differentiate between internal and external quality attributes. ★★★ (Slide 7)

Internal Attribute	External Attribute
Derived from the product or process **description**	Derived from the relationship between the system and its **environment**
The system does **not** need to run	The system **must run**
Example: Efficiency	Example: Reliability, Robustness

Internal quality **leads to** external quality.

Q8. What is a quality metric? ★★★ (Slide 8–9)

To know how good or bad a product is, we need **measurement**. For this, we define the quality we want to measure and agree on a **scale**, called a metric.

A metric is a **qualitative or quantitative measurement scale** for a quality attribute, together with a **method** to measure it.

- **Internal metrics:** complexity of code logic, performance of specific functions
- **External metrics:** tests of the system's functions, reliability and performance

Q9. Differentiate between correctness and reliability. ★★★ (Slide 24–25)

Correctness	Reliability
System behaves **according to its specification**	**Probability** that the system works as expected over a period of time
Absolute property	**Relative** property
A system **cannot be "almost correct"**	Example: mean time between failures is 3 weeks

Q10. What is a quality attribute? ★★ (Slide 5–6)

A quality attribute is a **property of a process or product** that has a qualitative or quantitative value and can be **measured or observed**.

Examples: maintainability, portability, functionality, usability.

Quality attributes apply to both the **product** (delivered to the customer) and the **process** (that produces the product).

Q11. Define software quality. ★★ (Slide 2)

Software quality is conformance to:

1. explicitly stated **functional and performance requirements**,
2. explicitly documented **development standards**, and
3. **implicit characteristics** that are expected of all professionally developed software.

Example: a banking app must do correct transactions (functional), follow coding standards (documented), and be secure and easy to use (implicit).

Q12. What are the barriers to achieving quality? ★★ (Slide 10–11)

Common reasons for not achieving high quality are:

- Misunderstanding the **importance** of quality attributes
- Poor **languages** to write quality requirements
- Poor **modelling methods** to design for quality
- **Difficulty** in designing for quality attributes
- Lack of documented **design patterns**
- **Quality control as an afterthought** in most projects

Q13. Differentiate between availability and recoverability. ★★ (Slide 32–33)

Availability	Recoverability
Proportion of time the system is **up and running**	Ability of the system to **start again from where it stopped** after a crash
Related to **scalability**	May run **diagnostic programs** before restarting
Example: a website is available 99% of the time	Example: MS Word recovers your document after a crash

Q14. What is maintainability? ★★ (Slide 39–41)

Maintainability is how easy it is to **change** a system after its first release.

Sub-qualities: Analysability, Changeability, Stability, Testability.

It also refers to:

- **Repairability:** work needed to fix a defect
- **Evolvability:** work needed to adapt to new requirements
- **Portability:** work needed to move to a new platform

Q15. What are the process quality attributes? ★★ (Slide 47–49)

- **Productivity:** amount of product made with a given amount of resources
- **Timeliness:** ability to deliver the product **on time**; important for marketing ("short time to market")
- **Visibility (Transparency):** process steps and project status are **clear and accessible** to everyone

How to write a short answer

- Start with a clear **definition**.
- If the question says "differentiate", draw a **table**.
- Add **one example**.
- **Underline** the keywords.